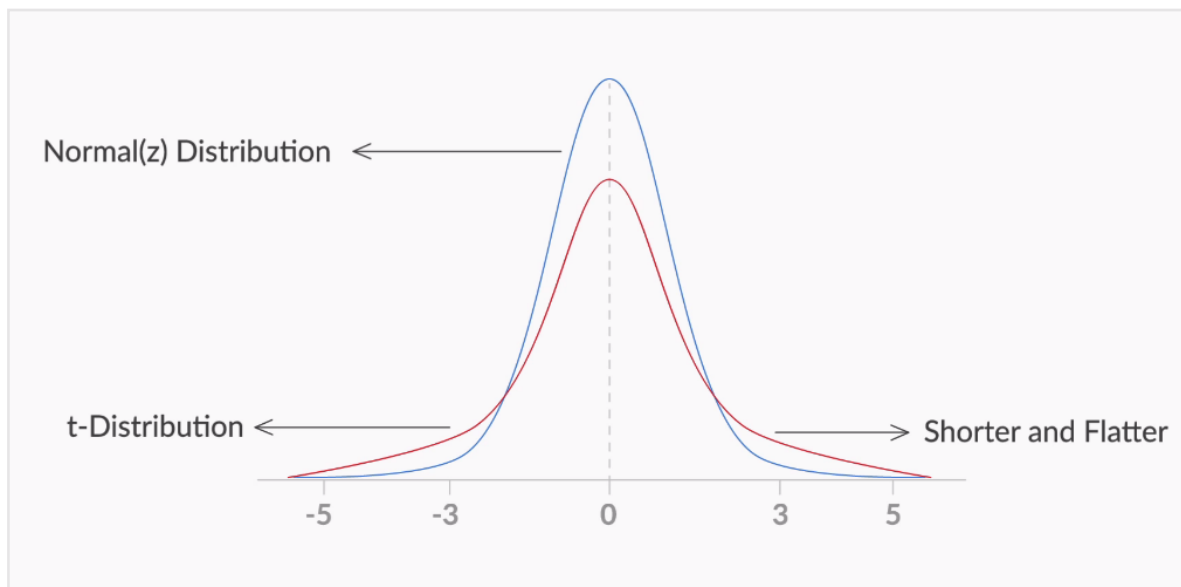
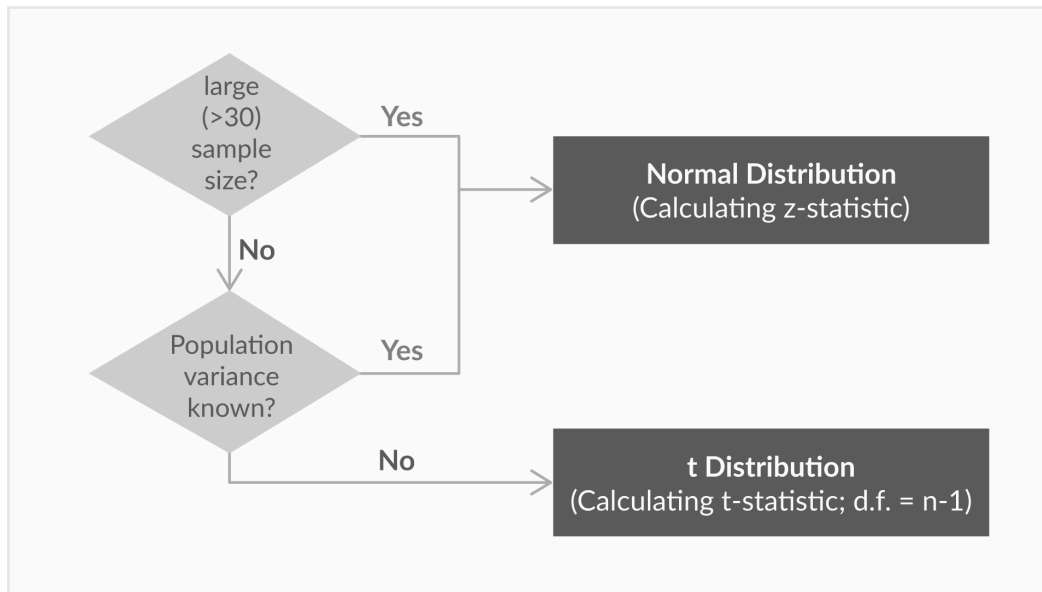


A t-distribution is also referred to as **Student's T distribution**. A t-distribution is similar to the normal distribution in many cases; for example, it is symmetrical about its central tendency. However, it is shorter than the normal distribution and has a flatter tail, which would eventually mean that it has a larger standard deviation.



At a sample size beyond 30, the t-distribution becomes approximately equal to the normal distribution.

The most important use of the t-distribution is that you can approximate the value of the **standard deviation of the population ( $\sigma$ )** from the **sample standard deviation (s)**. However, as the sample size increases more than 30, the t-value tends to be equal to the z-value. Thus, if you want to summarise the decision-making in a flowchart, this is what you would get.



Look at the flowchart above and answer the following questions.

## T Distribution - Part 2

Let's look at how the method of **making a decision** changes if you are using the sample's standard deviation instead of the population's. If you recall the critical value method, the first step is as follows:

1. Calculate the value of  $Z_c$  from the given value of  $\alpha$  (significance level). Take it as 5% if not specified in the problem.

So, to find  $Z_c$ , you would use the **t-table** instead of the z-table. The **t-table** contains values of  $Z_c$  for a given degree of freedom and value of  $\alpha$  (significance level).  $Z_c$ , in this case, can also be called as t-statistic (critical).

Download the t-table given ahead and attempt the following questions to understand how to use the t-table to find  $Z_c$ .

## T-table

In the second question, you used the t-table to find the value of  $Z_c$  for sample size = 32 and a significance level of 5%. If you use the z-table for the same, you would get the same value of  $Z_c$ , since, **for sample size  $\geq 30$ , the t-distribution is the same as the z-distribution.**

Practically you would not need to refer to the z-table or t-table when doing hypothesis testing in the industry. Going forward when you need to do hypothesis testing in demonstrations of Excel or R, you would use the term **t-test** since that is mostly performed in the industry. All calculations and results of a t-test are same as the z-test whenever the sample size  $\geq 30$ .

Before you proceed further, spend some time answering the question next.

A critical-T value is a “cut-off point” on the t distribution.

**A t-distribution is a probability distribution that is used to calculate population parameters when the sample size is small and when the population variance is unknown. T values are used to analyze whether to support or reject a null hypothesis.**

After conducting a t-test, we get its statistics as result. In order to determine the significance of the result, we compare the t- score obtained by a critical t value. If the absolute value of the t-score is greater than the t critical value, then the results of the test are statistically significant.

### **Formula:**

$$t = [ \bar{x} - \mu ] / [ s / \sqrt{n} ]$$

where,

- $t$  = t score

- $\bar{x}$  = sample mean,
- $\mu$  = population mean,
- $s$  = standard deviation of the sample,
- $n$  = sample size

While developing an e-commerce website, there could be different opinions about the choices of various elements, such as the shape of buttons, the text on the call-to-action buttons, the colour of various UI elements, the copy on the website, or numerous other such things.

Often, the choice of these elements is very subjective and is difficult to predict which option would perform better. To resolve such conflicts, you can use A/B testing. **A/B testing** provides a way for you to test two different versions of the same element and see which one performs better. You can read more about A/B testing from this [link](#).

You can see a few more case studies and applications of A/B testing in the real world [here](#).

Hypothesis testing in Python

## **Hypothesis testing in Python**

You can easily do hypothesis testing in Python by using **stats** from **Scipy** library.

### **1-sample t-test: testing the value of a population mean**

To test, if the population mean of data is likely to be equal to a given value

`scipy.stats.ttest_1samp()`

`stats.ttest_1samp(data['column'], x)`

#where x is the mean value you want to test

## **2-sample t-test: testing for difference across populations**

```
scipy.stats.ttest_ind()
```

```
stats.ttest_ind(column_1,column_2)
```

## **Paired tests: repeated measurements on the same individuals**

```
stats.ttest_rel()
```

```
stats.ttest_rel(column_1,column_2)
```

Hypothesis testing in Python

## **Hypothesis testing in Python**

You can easily do hypothesis testing in Python by using **stats** from **Scipy** library.

## **1-sample t-test: testing the value of a population mean**

To test, if the population mean of data is likely to be equal to a given value

```
scipy.stats.ttest_1samp()
```

```
stats.ttest_1samp(data['column'], x)
```

#where x is the mean value you want to test

## **2-sample t-test: testing for difference across populations**

```
scipy.stats.ttest_ind()
```

```
stats.ttest_ind(column_1,column_2)
```

## **Paired tests: repeated measurements on the same individuals**

```
stats.ttest_rel()
```

```
stats.ttest_rel(column_1,column_2)
```

To summarise, hypothesis testing still holds importance in the following two types of industries, even if all the data is available digitally:

1. Manufacturing processes in the food, pharmaceuticals, chemicals industries, where it is not practically possible to gather information on the entire population.
2. E-commerce, advertising and digital marketing companies, where the amount of data collected from various samples is so huge that analysing all data becomes very difficult without having big data systems in place.

Summary

## Summary

So what did you learn in this session?

### 1. **T-distribution:**

- A T-distribution is used whenever the standard deviation of the population is unknown
- The degrees of freedom of a T-distribution is equal to sample size  $n - 1$
- For sample size  $\geq 30$ , the T-distribution becomes the same as the normal distribution
- The output values and results of both t-test and z-test are same for sample size  $\geq 30$

### 2. **Two-sample mean test - paired:**

- It is used when your sample observations are from the same individual or object
- During this test, you are testing the same

subject twice

### **3. Two-sample mean test - unpaired:**

- During this test, you are not testing the same subject twice
- It is used when your sample observations are independent

### **4. Two-sample proportion test:**

- It is used when your sample observations are categorical, with two categories
- It could be True/False, 1/0, Yes/No, Male/Female, Success/Failure, etc.

### **5. A/B Testing:**

- A/B testing is a direct industry application of the two-sample proportion test
- It is a widely used process in digital companies in the ecommerce, manufacturing and advertising domains
- It provides a way to test two different versions of the same element and see which one performs better

You can download the lecture notes for the module in the next section. The lecture notes include a summary of the entire module.

[Lecture Notes - Hypothesis Testing](#)

**Note :** The number of stores (page number 6 or page which contains Figure 5) should be 36 instead of 25.

